

JEE (MAIN+ADVANCED)

Mathematics

ENTHUSIAST COURSE



STUDY MATERIAL

Methods of Differentiation

English Medium

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METHODS OF DIFFERENTIATION

The process of calculating derivative is called differentiation.

1. DERIVATIVE OF $f(x)$ FROM THE FIRST PRINCIPLE :

Obtaining the derivative using the definition $\lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} = \lim_{\delta x \rightarrow 0} \frac{f(x + \delta x) - f(x)}{\delta x} = f'(x) = \frac{dy}{dx}$ is called calculating derivative using first principle or ab initio or delta method.

Note : $\frac{dy}{dx}$ can also be represented as y_1 or y' or Dy or $f'(x)$. $\frac{dy}{dx}$ represents instantaneous rate of change of y w.r.t. x .

Illustration 1 : Differentiate each of following functions by first principle :

(i) $f(x) = \tan x$

(ii) $f(x) = e^{\sin x}$

Solution :

$$\begin{aligned} \text{(i)} \quad f'(x) &= \lim_{h \rightarrow 0} \frac{\tan(x+h) - \tan x}{h} = \lim_{h \rightarrow 0} \frac{\tan(x+h-x)[1 + \tan x \tan(x+h)]}{h} \\ &= \lim_{h \rightarrow 0} \frac{\tanh}{h} \cdot (1 + \tan^2 x) = \sec^2 x. \end{aligned}$$

Ans.

$$\begin{aligned} \text{(ii)} \quad f'(x) &= \lim_{h \rightarrow 0} \frac{e^{\sin(x+h)} - e^{\sin x}}{h} = \lim_{h \rightarrow 0} e^{\sin x} \frac{[e^{\sin(x+h) - \sin x} - 1]}{\sin(x+h) - \sin x} \left(\frac{\sin(x+h) - \sin x}{h} \right) \\ &= e^{\sin x} \lim_{h \rightarrow 0} \frac{\sin(x+h) - \sin x}{h} = e^{\sin x} \cos x \end{aligned}$$

Ans.

Do yourself -1 :

1. Differentiate each of following functions by first principle:

(a) $f(x) = \ln x$

(b) $f(x) = \frac{1}{x}$

(c) $\tan \sqrt{x}$

(d) $\cos(\sin x)$

(e) $\cos^{-1} x$

(f) $\sec^{-1} x$

2. DERIVATIVE OF STANDARD FUNCTIONS :

	f(x)	f'(x)		f(x)	f'(x)
(i)	x^n	nx^{n-1}	(ii)	e^x	e^x
(iii)	a^x	$a^x \ln a, a > 0$	(iv)	$\ln x$	$1/x$
(v)	$\log_a x$	$(1/x) \log_a e, a > 0, a \neq 1$	(vi)	$\sin x$	$\cos x$
(vii)	$\cos x$	$-\sin x$	(viii)	$\tan x$	$\sec^2 x$
(ix)	$\sec x$	$\sec x \tan x$	(x)	$\operatorname{cosec} x$	$-\operatorname{cosec} x \cdot \cot x$
(xi)	$\cot x$	$-\operatorname{cosec}^2 x$	(xii)	constant	0
(xiii)	$\sin^{-1} x$	$\frac{1}{\sqrt{1-x^2}}, -1 < x < 1$	(xiv)	$\cos^{-1} x$	$\frac{-1}{\sqrt{1-x^2}}, -1 < x < 1$
(xv)	$\tan^{-1} x$	$\frac{1}{1+x^2}, x \in \mathbb{R}$	(xvi)	$\sec^{-1} x$	$\frac{1}{ x \sqrt{x^2-1}}, x > 1$
(xvii)	$\operatorname{cosec}^{-1} x$	$\frac{-1}{ x \sqrt{x^2-1}}, x > 1$	(xviii)	$\cot^{-1} x$	$\frac{-1}{1+x^2}, x \in \mathbb{R}$

3. FUNDAMENTAL THEOREMS :

If f and g are derivable functions of x, then,

$$(a) \quad \frac{d}{dx}(f \pm g) = \frac{df}{dx} \pm \frac{dg}{dx}$$

$$(b) \quad \frac{d}{dx}(cf) = c \frac{df}{dx}, \text{ where } c \text{ is any constant}$$

$$(c) \quad \frac{d}{dx}(fg) = f \frac{dg}{dx} + g \frac{df}{dx} \text{ known as "PRODUCT RULE"}$$

$$(d) \quad \frac{d}{dx}\left(\frac{f}{g}\right) = \frac{g\left(\frac{df}{dx}\right) - f\left(\frac{dg}{dx}\right)}{g^2} \text{ where } g \neq 0 \text{ known as "QUOTIENT RULE"}$$

$$(e) \quad \text{If } y = f(u) \text{ \& } u = g(x) \text{ then } \frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} \text{ known as "CHAIN RULE"}$$

Note : In general if $y = f(u)$ then $\frac{dy}{dx} = f'(u) \cdot \frac{du}{dx}$.

Illustration 2 : If $y = e^x \tan x + x \log_e x$, find $\frac{dy}{dx}$.

Solution :

$$y = e^x \cdot \tan x + x \cdot \log_e x$$

On differentiating we get,

$$\frac{dy}{dx} = e^x \cdot \tan x + e^x \cdot \sec^2 x + 1 \cdot \log x + x \cdot \frac{1}{x}$$

$$\text{Hence, } \frac{dy}{dx} = e^x(\tan x + \sec^2 x) + (\log x + 1)$$

Ans.

Illustration 3 : If $y = \frac{\log x}{x} + e^x \sin 2x + \log_5 x$, find $\frac{dy}{dx}$.

Solution : On differentiating we get,

$$\frac{dy}{dx} = \frac{d}{dx} \left(\frac{\log x}{x} \right) + \frac{d}{dx} (e^x \sin 2x) + \frac{d}{dx} (\log_5 x) = \frac{\frac{1}{x} \cdot x - \log x \cdot 1}{x^2} + e^x \sin 2x + 2e^x \cdot \cos 2x + \frac{1}{x \log_e 5}$$

Hence, $\frac{dy}{dx} = \left(\frac{1 - \log x}{x^2} \right) + e^x (\sin 2x + 2 \cos 2x) + \frac{1}{x \log_e 5}$ **Ans.**

Illustration 4 : If $y = \log_e (\tan^{-1} \sqrt{1+x^2})$, find $\frac{dy}{dx}$.

Solution : $y = \log_e (\tan^{-1} \sqrt{1+x^2})$

On differentiating we get,

$$\begin{aligned} &= \frac{1}{\tan^{-1} \sqrt{1+x^2}} \cdot \frac{1}{1+(\sqrt{1+x^2})^2} \cdot \frac{1}{2\sqrt{1+x^2}} \cdot 2x \\ &= \frac{x}{(\tan^{-1} \sqrt{1+x^2}) \{1+(\sqrt{1+x^2})^2\} \sqrt{1+x^2}} = \frac{x}{(\tan^{-1} \sqrt{1+x^2}) (2+x^2) \sqrt{1+x^2}} \end{aligned}$$
 Ans.

Do yourself -2 :

1. Find $\frac{dy}{dx}$ if -

(a) $y = (x+1)(x+2)(x+3)$

(b) $y = e^{5x} \tan(x^2 + 2)$

2. $\frac{d}{dx} \left(\sqrt{x} + \frac{1}{\sqrt{x}} \right)^2 =$

(A) $1 - \frac{1}{x^2}$

(B) $1 + \frac{1}{x^2}$

(C) $1 - \frac{1}{2x}$

(D) None of these

3. If $y = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \infty$, then $\frac{dy}{dx} =$

(A) y

(B) $y - 1$

(C) $y + 1$

(D) None of these

4. If $f(x) = x \tan^{-1} x$, then $f'(1) =$

(A) $1 + \frac{\pi}{4}$

(B) $\frac{1}{2} + \frac{\pi}{4}$

(C) $\frac{1}{2} - \frac{\pi}{4}$

(D) 2

5. If $f(x) = \log_x (\log x)$, then $f'(x)$ at $x = e$ is

(A) e

(B) $\frac{1}{e}$

(C) 1

(D) None of these

6. $\frac{d}{dx}(e^x \log \sin 2x) =$
 (A) $e^x (\log \sin 2x + 2 \cot 2x)$ (B) $e^x (\log \cos 2x + 2 \cot 2x)$
 (C) $e^x (\log \cos 2x + \cot 2x)$ (D) None of these
7. If $y = \sin(\sqrt{\sin x + \cos x})$, then $\frac{dy}{dx} =$
 (A) $\frac{1}{2} \frac{\cos \sqrt{\sin x + \cos x}}{\sqrt{\sin x + \cos x}}$ (B) $\frac{\cos \sqrt{\sin x + \cos x}}{\sqrt{\sin x + \cos x}}$
 (C) $\frac{1}{2} \frac{\cos \sqrt{\sin x + \cos x}}{\sqrt{\sin x + \cos x}} \cdot (\cos x - \sin x)$ (D) None of these
8. If $y = \sec x^\circ$, then $dy/dx =$
 (A) $\sec x \tan x$ (B) $\sec x^\circ \tan x^\circ$
 (C) $\frac{\pi}{180} \sec x^\circ \tan x^\circ$ (D) $\frac{180}{\pi} \sec x^\circ \tan x^\circ$
9. The derivative of $f(x) = |x^2 - x|$ at $x = 2$ is
 (A) -3 (B) 0 (C) 3 (D) Not defined
10. $\frac{d}{dx} \left[\left(\frac{\tan^2 2x - \tan^2 x}{1 - \tan^2 2x \tan^2 x} \right) \cot 3x \right]$
 (A) $\tan 2x \tan x$ (B) $\tan 3x \tan x$ (C) $\sec^2 x$ (D) $\sec x \tan x$
11. If $y = \frac{1}{1+x^{\beta-\alpha}+x^{\gamma-\alpha}} + \frac{1}{1+x^{\alpha-\beta}+x^{\gamma-\beta}} + \frac{1}{1+x^{\alpha-\gamma}+x^{\beta-\gamma}}$, then $\frac{dy}{dx}$ is equal to -
 (A) 0 (B) 1
 (C) $(\alpha + \beta + \gamma)x^{\alpha+\beta+\gamma-1}$ (D) $\alpha\beta\gamma$
12. If $f(x) = |\cos x|$, then $f'\left(\frac{3\pi}{4}\right)$ is equal to -
 (A) $-\frac{1}{\sqrt{2}}$ (B) $\frac{1}{\sqrt{2}}$ (C) 1 (D) -1
13. $\frac{d}{dx}(e^x \sin \sqrt{3} x)$ equals -
 (A) $e^x \sin(\sqrt{3} x + \pi/3)$ (B) $2e^x \sin(\sqrt{3} x + \pi/3)$
 (C) $\frac{1}{2} e^x \sin(\sqrt{3} x + \pi/3)$ (D) $\frac{1}{2} e^x \sin(\sqrt{3} x - \pi/3)$
14. $\frac{d}{dx}(\ln \sin \sqrt{x})$ is equal to -
 (A) $\frac{\tan \sqrt{x}}{2\sqrt{x}}$ (B) $\frac{\cot \sqrt{x}}{\sqrt{x}}$ (C) $\frac{\cot \sqrt{x}}{2x}$ (D) $\frac{\cot \sqrt{x}}{2\sqrt{x}}$

Differentiate the given functions with respect to x :

15. $y = \frac{x}{\sin x + \cos x}$

16. $y = \frac{1}{3} \tan^3 x - \tan x + x$

17. $y = x \sec^2 x - \tan x$

18. $y = a \cos \frac{x}{3}$

19. $y = \tan \frac{x+1}{2}$

20. $y = \sqrt{1 + 2 \tan x}$

21. $y = \cos^3 4x$

22. $y = \sqrt{\tan \frac{x}{2}}$

23. $y = \sin \sqrt{1+x^2}$

24. $y = \cot \sqrt[3]{1+x^2}$

25. $y = (1 + \sin^2 x)^4$

26. $y = \sqrt{1 + \tan \left(x + \frac{1}{x} \right)}$

27. $y = \cos^2 \frac{1-\sqrt{x}}{1+\sqrt{x}}$

28. $y = \sin^2 (\cos 3x)$

29. $y = x \arcsin x$

30. $y = \frac{\arcsin x}{\arccos x}$

31. $y = (\arcsin x)^2$

32. $y = \frac{1}{\arcsin x}$

33. $y = x \sin x \arctan x$

34. $y = (\arccos x + \arcsin x)^n$

35. $y = \frac{x}{1+x^2} - \arctan x$

36. $y = \arctan x^2$

37. $y = \frac{1}{2} \sqrt[4]{\arcsin \sqrt{x^2 + 2x}}$

38. $y = \arccos \left(\frac{b + a \cos x}{a + b \cos x} \right); (a, b > 0, \sin x > 0)$

39. $y = x^2 \log_3 x$

40. $y = \sqrt{\ell n x}$

41. $y = x \sin x \ell n x$

42. $y = \frac{\ell n x}{x^n}$

43. $y = \frac{1 - \ell n x}{1 + \ell n x}$

44. $y = \ell n(x^2 - 4x)$

45. $y = \ell n \tan x$

46. $y = \log_2 [\log_3 (\log_5 x)]$

47. $y = \ell n \arctan \sqrt{1+x^2}$

48. $y = \sqrt[3]{\ell n \sin \frac{x+3}{4}}$

49. $y = x \cdot 10^x$

50. $y = \frac{x^3 + 2^x}{e^x}$

51. $y = \frac{\cos x}{e^x}$

52. $y = 2^{\frac{x}{\ell n x}}$

53. $y = \frac{1-10^x}{1+10^x}$

54. $y = 3^{\sin x}$

55. $y = ae^{-b^2 x^2}$

56. $y = Ae^{-k^2 x} \sin(\omega x + \alpha)$

57. Prove that the function $y = \ell n \left(\frac{1}{1+x} \right)$ satisfies the relationship $xy' + 1 = e^y$ (where dash denotes derivative)

4. LOGARITHMIC DIFFERENTIATION :

To find the derivative of a function :

- (a) which is the product or quotient of a number of functions or
(b) of the form $[f(x)]^{g(x)}$ where f & g are both derivable functions.

It is convenient to take the logarithm of the function first & then differentiate.

Illustration 5 : If $y = (\sin x)^{\csc x}$, find $\frac{dy}{dx}$

Solution : $\ln y = \ln x \cdot \ln (\sin x)$
On differentiating we get,

$$\frac{1}{y} \frac{dy}{dx} = \frac{1}{x} \ln(\sin x) + \ln x. \quad \frac{\cos x}{\sin x} \Rightarrow \frac{dy}{dx} = (\sin x)^{\ln x} \left[\frac{\ln(\sin x)}{x} + \cot x \ln x \right] \quad \text{Ans.}$$

Illustration 6: If $x = \exp \left(\tan^{-1} \left(\frac{y - x^2}{x^2} \right) \right)$, then $\frac{dy}{dx}$ equals -

- (A) $x [1 + \tan (\log x) + \sec^2 x]$ (B) $2x [1 + \tan (\log x)] + \sec^2 x$
(C) $2x [1 + \tan (\log x)] + \sec x$ (D) $2x + x[1 + \tan(\log x)]^2$

Solution : Taking log on both sides, we get

$$\log x = \tan^{-1} \left(\frac{y - x^2}{x^2} \right) \Rightarrow \tan(\log x) = (y - x^2) / x^2$$

$$\Rightarrow y = x^2 + x^2 \tan(\log x)$$

On differentiating, we get

$$\therefore \frac{dy}{dx} = 2x + 2x \tan(\log x) + x \sec^2(\log x) \Rightarrow 2x [1 + \tan(\log x)] + x \sec^2(\log x)$$
$$= 2x + x[1 + \tan(\log x)]^2 \quad \text{Ans. (D)}$$

Illustration 7 : If $y = \frac{x^{1/2}(1-2x)^{2/3}}{(2-3x)^{3/4}(3-4x)^{4/5}}$ find $\frac{dy}{dx}$

Solution : $\ell_n y = \frac{1}{2} \ell_n x + \frac{2}{3} \ell_n (1 - 2x) - \frac{3}{4} \ell_n (2 - 3x) - \frac{4}{5} \ell_n (3 - 4x)$

On differentiating we get,

$$\Rightarrow \frac{1}{y} \frac{dy}{dx} = \frac{1}{2x} - \frac{4}{3(1-2x)} + \frac{9}{4(2-3x)} + \frac{16}{5(3-4x)}$$

$$\frac{dy}{dx} = y \left(\frac{1}{2x} - \frac{4}{3(1-2x)} + \frac{9}{4(2-3x)} + \frac{16}{5(3-4x)} \right) \quad \text{Ans.}$$

Do yourself -3 :

1. Find $\frac{dy}{dx}$ if $y = x^x$
2. Find $\frac{dy}{dx}$ if $y = e^x \cdot e^{x^2} \cdot e^{x^3} \cdot e^{x^4}$
3. Let $f(x) = (x-1)^{100} (x-2)^{2(99)} (x-3)^{3(98)} \dots (x-100)^{100}$, then the value of $\frac{f'(101)}{f(101)}$ is
4. If $y = e^{x+e^{x+e^{x+\dots\infty}}}$, then $\frac{dy}{dx} =$
 - (A) $\frac{y}{1-y}$
 - (B) $\frac{1}{1-y}$
 - (C) $\frac{y}{1+y}$
 - (D) $\frac{y}{y-1}$
5. If $y = (x^x)^x$, then $\frac{dy}{dx} =$
 - (A) $(x^x)^x (1 + 2 \log x)$
 - (B) $(x^x)^x (1 + \log x)$
 - (C) $x(x^x)^x (1 + 2 \log x)$
 - (D) $x(x^x)^x (1 + \log x)$
6. If $y = x^{\sin x}$, then $\frac{dy}{dx} =$
 - (A) $\left(\frac{x \cos x \cdot \log x + \sin x}{x} \right) \cdot x^{\sin x}$
 - (B) $\frac{y[x \cos x \cdot \log x + \cos x]}{x}$
 - (C) $y[x \sin x \cdot \log x + \cos x]$
 - (D) None of these
7. $\frac{d}{dx} \{(\sin x)^x\} =$
 - (A) $\left[\frac{x \cos x + \sin x \log \sin x}{\sin x} \right]$
 - (B) $(\sin x)^x \left[\frac{x \cos x + \sin x \log \sin x}{\sin x} \right]$
 - (C) $(\sin x)^x \left[\frac{x \sin x + \sin x \log \sin x}{\sin x} \right]$
 - (D) None of these
8. $\frac{d}{dx} \{(\sin x)^{\log x}\} =$
 - (A) $(\sin x)^{\log x} \left[\frac{1}{x} \log \sin x + \cot x \right]$
 - (B) $(\sin x)^{\log x} \left[\frac{1}{x} \log \sin x + \cot x \log x \right]$
 - (C) $(\sin x)^{\log x} \left[\frac{1}{x} \log \sin x + \log x \right]$
 - (D) None of these

Differentiate the given functions with respect to x :

9. $y = x^{x^x}$
10. $y = (\sin x)^{\cos x}$
11. $y = (\ell n x)^x$
12. $y = x^{\ell n x}$
13. $y = \frac{(x+1)^3 \sqrt[4]{x-2}}{\sqrt[5]{(x-3)^2}}$
14. $y = \sqrt{x \sin x} \sqrt{1-e^x}$
15. $y = x^{\frac{1}{x}}$
16. $y = 2x^{\sqrt{x}}$
17. $y = (x^2 + 1)^{\sin x}$
18. $y = \sqrt[3]{\frac{x(x^2+1)}{(x^2-1)^2}}$

5. PARAMETRIC DIFFERENTIATION :

If $y = f(\theta)$ & $x = g(\theta)$ where θ is a parameter, then $\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{f'(\theta)}{g'(\theta)}$

Illustration 8: If $y = a \cos t$ and $x = a(t - \sin t)$ find the value of $\frac{dy}{dx}$ at $t = \frac{\pi}{2}$

Solution : $\frac{dy}{dx} = \frac{-a \sin t}{a(1 - \cos t)} \Rightarrow \frac{dy}{dx} \bigg|_{t=\frac{\pi}{2}} = -1$

Ans.

Illustration 9 : Prove that the function represented parametrically by the equations. $x = \frac{1+t}{t^3}$; $y = \frac{3}{2t^2} + \frac{2}{t}$

satisfies the relationship : $x(y')^3 = 1 + y'$ (where $y' = \frac{dy}{dx}$)

Solution : Here $x = \frac{1+t}{t^3} = \frac{1}{t^3} + \frac{1}{t^2}$

Differentiating w.r. to t

$$\frac{dx}{dt} = -\frac{3}{t^4} - \frac{2}{t^3}$$

$$y = \frac{3}{2t^2} + \frac{2}{t}$$

Differentiating w.r. to t

$$\frac{dy}{dt} = -\frac{3}{t^3} - \frac{2}{t^2}$$

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = t = y'$$

Since $x = \frac{1+t}{t^3} \Rightarrow x = \frac{1+y'}{(y')^3}$ or $x(y')^3 = 1 + y'$

Ans.

6. DERIVATIVE OF A FUNCTION W.R.T. ANOTHER FUNCTION :

Let $y = f(x)$; $z = g(x)$ then $\frac{dy}{dz} = \frac{dy/dx}{dz/dx} = \frac{f'(x)}{g'(x)}$

Illustration 10: Differentiate $\log_e(\tan x)$ with respect to $\sin^{-1}(e^x)$.

Solution : $\frac{d(\log_e \tan x)}{d(\sin^{-1}(e^x))} = \frac{\frac{d}{dx}(\log_e \tan x)}{\frac{d}{dx} \sin^{-1}(e^x)} = \frac{\cot x \cdot \sec^2 x}{e^x \cdot 1/\sqrt{1-e^{2x}}} = \frac{e^{-x} \sqrt{1-e^{2x}}}{\sin x \cos x}$

Ans.

Do yourself-4 :

- Find $\frac{dy}{dx}$ at $t = \frac{\pi}{4}$ if $y = \cos^4 t$ & $x = \sin^4 t$.
- Find the slope of the tangent at a point P(t) on the curve $x = at^2$, $y = 2at$.
- Differentiate $x^{\ell nx}$ with respect to ℓnx .
- If $x = a \left(\cos t + \log \tan \frac{t}{2} \right)$, $y = a \sin t$, then $\frac{dy}{dx} =$
 (A) $\tan t$ (B) $-\tan t$ (C) $\cot t$ (D) $-\cot t$
- If $x = \frac{1-t^2}{1+t^2}$ and $y = \frac{2t}{1+t^2}$, then $\frac{dy}{dx} =$
 (A) $\frac{-y}{x}$ (B) $\frac{y}{x}$ (C) $\frac{-x}{y}$ (D) $\frac{x}{y}$
- If $y = \frac{1}{4}u^4$, $u = \frac{2}{3}x^3 + 5$, then $\frac{dy}{dx} =$
 (A) $\frac{1}{27}x^2(2x^3 + 15)^3$ (B) $\frac{2}{27}x(2x^3 + 5)^3$
 (C) $\frac{2}{27}x^2(2x^3 + 15)^3$ (D) None of these
- If $x = \sin^{-1}(3t - 4t^3)$ and $y = \cos^{-1}\sqrt{1-t^2}$, then $\frac{dy}{dx}$ is equal to
 (A) $\frac{1}{2}$ (B) $\frac{2}{5}$ (C) $\frac{3}{2}$ (D) $\frac{1}{3}$
- If $x = a(t - \sin t)$, $y = a(1 + \cos t)$, then $\frac{dy}{dx}$ equals-
 (A) $-\tan \frac{t}{2}$ (B) $\cot \frac{t}{2}$ (C) $-\cot \frac{t}{2}$ (D) $\tan \frac{t}{2}$

7. DIFFERENTIATION OF IMPLICIT FUNCTIONS : $\phi(x, y) = 0$

- (a) To find dy/dx of implicit functions, we differentiate each term w.r.t. x regarding y as a function of x & then collect terms with dy/dx together on one side.

- (b) Also $\frac{dy}{dx} = -\frac{\frac{\partial \phi}{\partial x}}{\frac{\partial \phi}{\partial y}}$, where $\frac{\partial \phi}{\partial x}$ = partial derivative of $\phi(x, y)$ w.r.t. x taking y as a constant and

$\frac{\partial \phi}{\partial y}$ = partial derivative of $\phi(x, y)$ w.r.t. y taking x as a constant.

- (c) In the case of implicit functions, generally, both x & y are present in answers of dy/dx .

Illustration 11 : If $x^y + y^x = 2$, then find $\frac{dy}{dx}$.

Solution : Let $u = x^y$ and $v = y^x$
 $u + v = 2 \Rightarrow \frac{du}{dx} + \frac{dv}{dx} = 0$
 Now $u = x^y$ and $v = y^x$
 $\Rightarrow \ln u = y \ln x$ and $\ln v = x \ln y$
 $\Rightarrow \frac{1}{u} \frac{du}{dx} = \frac{y}{x} + \ln x \frac{dy}{dx}$ and $\frac{1}{v} \frac{dv}{dx} = \ln y + \frac{x}{y} \frac{dy}{dx}$
 $\Rightarrow \frac{du}{dx} = x^y \left(\frac{y}{x} + \ln x \frac{dy}{dx} \right)$ and $\frac{dv}{dx} = y^x \left(\ln y + \frac{x}{y} \frac{dy}{dx} \right)$
 $\Rightarrow x^y \left(\frac{y}{x} + \ln x \frac{dy}{dx} \right) + y^x \left(\ln y + \frac{x}{y} \frac{dy}{dx} \right) = 0 \Rightarrow \frac{dy}{dx} = - \frac{\left(y^x \ln y + x^y \cdot \frac{y}{x} \right)}{\left(x^y \ln x + y^x \cdot \frac{x}{y} \right)}$ **Ans.**

Aliter :

$$\phi(x, y) = x^y + y^x - 2 = 0$$

$$\frac{dy}{dx} = \frac{-\partial \phi / \partial x}{\partial \phi / \partial y} = \frac{yx^{y-1} + y^x \ln y}{x^y \ln x + xy^{x-1}}$$

Illustration 12 : If $y = \frac{\sin x}{1 + \frac{\cos x}{1 + \frac{\sin x}{1 + \cos x \dots}}}$, prove that $\frac{dy}{dx} = \frac{(1+y)\cos x + y \sin x}{1 + 2y + \cos x - \sin x}$.

Solution : Given function is $y = \frac{\sin x}{1 + \frac{\cos x}{1 + y}} = \frac{(1+y)\sin x}{1 + y + \cos x}$

$$\text{or } y + y^2 + y \cos x = (1 + y) \sin x \quad \dots(i)$$

Differentiate both sides with respect to x ,

$$\frac{dy}{dx} + 2y \frac{dy}{dx} + \frac{dy}{dx} \cos x - y \sin x = (1 + y) \cos x + \frac{dy}{dx} \sin x$$

$$\frac{dy}{dx} (1 + 2y + \cos x - \sin x) = (1 + y) \cos x + y \sin x$$

$$\text{or } \frac{dy}{dx} = \frac{(1+y)\cos x + y \sin x}{1 + 2y + \cos x - \sin x}$$
 Ans.

Aliter :

$$\text{From (i) } \phi(x, y) = (1 + y)\sin x - y - y^2 - y \cos x = 0$$

$$\frac{dy}{dx} = - \frac{\partial \phi / \partial x}{\partial \phi / \partial y} = - \frac{(1+y)\cos x + y \sin x}{\sin x - 1 - 2y - \cos x} = \frac{(1+y)\cos x + y \sin x}{1 + 2y + \cos x - \sin x}$$

Do yourself -5 :

- Find $\frac{dy}{dx}$, if $x + y = \sin(x - y)$
- If $x^2 + xe^y + y = 0$, find y' , also find the value of y' at point $(0,0)$.
- If $x\sqrt{1+y} + y\sqrt{1+x} = 0$, then $\frac{dy}{dx} =$
 (A) $1 + x$ (B) $(1 + x)^{-2}$ (C) $-(1 + x)^{-1}$ (D) $-(1 + x)^{-2}$
- If $\sin y = x \sin(a + y)$, then $\frac{dy}{dx} =$
 (A) $\frac{\sin^2(a + y)}{\sin(a + 2y)}$ (B) $\frac{\sin^2(a + y)}{\cos(a + 2y)}$ (C) $\frac{\sin^2(a + y)}{\sin a}$ (D) $\frac{\sin^2(a + y)}{\cos a}$
- If $2^x + 2^y = 2^{x+y}$, then $\frac{dy}{dx}$ is equal to -
 (A) $\frac{2^x + 2^y}{2^x - 2^y}$ (B) $\frac{2^x + 2^y}{1 + 2^{x+y}}$ (C) $2^{x-y} \left(\frac{2^y - 1}{1 - 2^x} \right)$ (D) $\frac{2^x + y - 2^x}{2^y}$
- If $\sin(xy) + \frac{x}{y} = x^2 - y$, then $\frac{dy}{dx} =$
 (A) $\frac{y[2xy - y^2 \cos(xy) - 1]}{xy^2 \cos(xy) + y^2 - x}$ (B) $\frac{[2xy - y^2 \cos(xy) - 1]}{xy^2 \cos(xy) + y^2 - x}$
 (C) $-\frac{y[2xy - y^2 \cos(xy) - 1]}{xy^2 \cos(xy) + y^2 - x}$ (D) None of these
- If $\sin^2 x + 2\cos y + xy = 0$, then $\frac{dy}{dx} =$
 (A) $\frac{y + 2\sin x}{2\sin y + x}$ (B) $\frac{y + \sin 2x}{2\sin y - x}$ (C) $\frac{y + 2\sin x}{\sin y + x}$ (D) None of these
- If $y = \sqrt{\log x + \sqrt{\log x + \sqrt{\log x + \dots \infty}}}$, then $\frac{dy}{dx} =$
 (A) $\frac{x}{2y - 1}$ (B) $\frac{x}{2y + 1}$ (C) $\frac{1}{x(2y - 1)}$ (D) $\frac{1}{x(1 - 2y)}$
- If $y = \sqrt{\sin x + \sqrt{\sin x + \sqrt{\sin x + \dots \infty}}}$, then
 (A) $(2y - 1)\frac{dy}{dx} - \sin x = 0$ (B) $(2y - 1)\cos x + \frac{dy}{dx} = 0$
 (C) $(2y - 1)\cos x - \frac{dy}{dx} = 0$ (D) $(2y - 1)\frac{dy}{dx} - \cos x = 0$

10. If $y = \sqrt{x + \sqrt{x + \sqrt{x + \dots \text{to } \infty}}}$ then $\frac{dy}{dx} =$
- (A) $\frac{x}{2y-1}$ (B) $\frac{2}{2y-1}$ (C) $\frac{-1}{2y-1}$ (D) $\frac{1}{2y-1}$
11. If $y = (\sin x)^{\tan x}$, then $\frac{dy}{dx}$ is equal to
- (A) $(\sin x)^{\tan x} (1 + \sec^2 x \log \sin x)$ (B) $\tan x (\sin x)^{\tan x - 1} \cos x$
 (C) $(\sin x)^{\tan x} \sec^2 x \log \sin x$ (D) $\tan x (\sin x)^{\tan x - 1}$
12. Let $f(x) = \cos 2x \cos 4x \cos 6x \cos 8x \cos 10x$ and $M = \lim_{x \rightarrow 0} \left(\frac{1 - (f(x))^3}{5 \tan^2 x} \right)$, where M is finite, then the value of $(\sqrt{M-2} + 1)$
13. If $f(x) = x + \frac{1}{2x + \frac{1}{2x + \frac{1}{2x + \dots \infty}}}$, then value of $f(2011) \cdot f'(2011)$ is
14. If $x^3 - y^3 + 3xy^2 - 3x^2y + 1 = 0$, then at $(0, 1)$ $\frac{dy}{dx}$ equals-
- (A) 1 (B) -1 (C) 2 (D) 0
15. If $y = \ln(x^{e^x \cdot a^x})^{y^x}$ find $\frac{dy}{dx}$.

8. DIFFERENTIATION BY TRIGONOMETRIC TRANSFORMATION :

Some Standard Substitutions :

Expression	Substitution
$\sqrt{a^2 - x^2}$	$x = a \sin \theta$ or $a \cos \theta$
$\sqrt{a^2 + x^2}$	$x = a \tan \theta$ or $a \cot \theta$
$\sqrt{x^2 - a^2}$	$x = a \sec \theta$ or $a \csc \theta$
$\sqrt{\frac{a+x}{a-x}}$ or $\sqrt{\frac{a-x}{a+x}}$	$x = a \cos \theta$ or $a \cos 2\theta$
$\sqrt{2ax - x^2}$	$x = a(1 - \cos \theta)$

Illustration 13: If $f(x) = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$ then find

- (i) $f(2)$ (ii) $f\left(\frac{1}{2}\right)$ (iii) $f(1)$

Solution : $x = \tan\theta$, where $-\frac{\pi}{2} < \theta < \frac{\pi}{2} \Rightarrow y = \sin^{-1}(\sin 2\theta)$

$$y = \begin{cases} \pi - 2\theta & \frac{\pi}{2} < 2\theta < \pi \\ 2\theta & -\frac{\pi}{2} \leq 2\theta \leq \frac{\pi}{2} \\ -(\pi + 2\theta) & -\pi < 2\theta < -\frac{\pi}{2} \end{cases} \Rightarrow f(x) = \begin{cases} \pi - 2 \tan^{-1} x & x > 1 \\ 2 \tan^{-1} x & -1 \leq x \leq 1 \\ -(\pi + 2 \tan^{-1} x) & x < -1 \end{cases}$$

$$\Rightarrow f(x) = \begin{cases} -\frac{2}{1+x^2} & x > 1 \\ \frac{2}{1+x^2} & -1 < x < 1 \\ \frac{-2}{1+x^2} & x < -1 \end{cases}$$

(i) $f(2) = -\frac{2}{5}$ (ii) $f\left(\frac{1}{2}\right) = \frac{8}{5}$ (iii) $f(1^+) = -1$ and $f(1^-) = +1 \Rightarrow f(1)$ does not exist **Ans.**

Illustration 14: $\frac{d}{dx} \left\{ \sin^2 \left(\cot^{-1} \sqrt{\frac{1+x}{1-x}} \right) \right\} =$

- (A) $-\frac{1}{2}$ (B) 0 (C) $\frac{1}{2}$ (D) -1

Solution : Let $y = \sin^2 \left(\cot^{-1} \sqrt{\frac{1+x}{1-x}} \right)$. Put $x = \cos 2\theta$ $\theta \in \left(0, \frac{\pi}{2} \right]$

$$\therefore y = \sin^2 \cot^{-1} \left(\sqrt{\frac{1+\cos 2\theta}{1-\cos 2\theta}} \right) = \sin^2 \cot^{-1} (\cot \theta)$$

$$\therefore y = \sin^2 \theta = \frac{1-\cos 2\theta}{2} = \frac{1-x}{2} = \frac{1}{2} - \frac{x}{2}$$

$$\therefore \frac{dy}{dx} = -\frac{1}{2}.$$

Ans (A)

Illustration 15: Obtain differential coefficient of $\tan^{-1} \frac{\sqrt{1+x^2}-1}{x}$ with respect to $\cos^{-1} \sqrt{\frac{1+\sqrt{1+x^2}}{2\sqrt{1+x^2}}}$

Solution : Assume $u = \tan^{-1} \frac{\sqrt{1+x^2}-1}{x}$, $v = \cos^{-1} \sqrt{\frac{1+\sqrt{1+x^2}}{2\sqrt{1+x^2}}}$

The function needs simplification before differentiation Let $x = \tan \theta$; $\theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

$$\therefore u = \tan^{-1} \left(\frac{\sec \theta - 1}{\tan \theta} \right) = \tan^{-1} \left(\frac{1 - \cos \theta}{\sin \theta} \right) = \tan^{-1} \left(\tan \frac{\theta}{2} \right) = \frac{\theta}{2}$$

$$v = \cos^{-1} \sqrt{\frac{1 + \sec \theta}{2 \sec \theta}} = \cos^{-1} \sqrt{\frac{1 + \cos \theta}{2}} = \cos^{-1} \left(\cos \frac{\theta}{2} \right) = \begin{cases} \frac{\theta}{2}, & \theta \in \left(0, \frac{\pi}{2}\right) \\ -\frac{\theta}{2}, & \theta \in \left(-\frac{\pi}{2}, 0\right) \end{cases}$$

$$\Rightarrow u = v \text{ or } -v$$

$$\Rightarrow \frac{du}{dv} = \begin{cases} 1, & x > 0 \\ -1, & x < 0 \end{cases}$$

Ans.

Do yourself : 6

1. If $y = \cos^{-1}(4x^3 - 3x)$, then find :

(a) $f' \left(-\frac{\sqrt{3}}{2} \right)$, (b) $f'(0)$, (c) $f' \left(\frac{\sqrt{3}}{2} \right)$

2. $\frac{d}{dx} \left(\tan^{-1} \frac{\cos x}{1 + \sin x} \right)$, $x \in (0, \pi)$ is equal to

(A) $-\frac{1}{2}$ (B) $\frac{1}{2}$ (C) -1 (D) 1

3. If $y = \sin^{-1} (x\sqrt{1-x} + \sqrt{x}\sqrt{1-x^2})$, $x \in \left(0, \frac{1}{2}\right)$ then $\frac{dy}{dx} =$

(A) $\frac{-2x}{\sqrt{1-x^2}} + \frac{1}{2\sqrt{x-x^2}}$ (B) $\frac{-1}{\sqrt{1-x^2}} - \frac{1}{2\sqrt{x-x^2}}$
 (C) $\frac{1}{\sqrt{1-x^2}} + \frac{1}{2\sqrt{x-x^2}}$ (D) None of these

4. $\frac{d}{dx} \left(\tan^{-1} \sqrt{\frac{1 + \cos \frac{x}{2}}{1 - \cos \frac{x}{2}}} \right), x \in \left(\frac{\pi}{2}, \pi \right)$ is equal to

(A) $-\frac{1}{4}$

(B) $\frac{1}{2}$

(C) $-\frac{1}{2}$

(D) $\frac{1}{4}$

5. If $y = \tan^{-1} \frac{4x}{1+5x^2} + \tan^{-1} \frac{2+3x}{3-2x}$, then $\frac{dy}{dx} =$

(A) $\frac{1}{1+25x^2} + \frac{2}{1+x^2}$

(B) $\frac{5}{1+25x^2} + \frac{2}{1+x^2}$

(C) $\frac{5}{1+25x^2}$

(D) $\frac{1}{1+25x^2}$

6. If $y = \tan^{-1} \left(\frac{x^{1/3} + a^{1/3}}{1 - x^{1/3} a^{1/3}} \right)$, then $\frac{dy}{dx} =$

(A) $\frac{1}{3x^{2/3} (1+x^{2/3})}$

(B) $\frac{1}{3x^{2/3} (1-x^{2/3})}$

(C) $-\frac{1}{3x^{2/3} (1+x^{2/3})}$

(D) $-\frac{a}{3x^{2/3} (1+x^{2/3})}$

7. If $y = \tan^{-1} \left(\frac{\sqrt{a} - \sqrt{x}}{1 + \sqrt{ax}} \right)$, then $\frac{dy}{dx} =$

(A) $\frac{1}{2(1+x)\sqrt{x}}$

(B) $\frac{1}{(1+x)\sqrt{x}}$

(C) $-\frac{1}{2(1+x)\sqrt{x}}$

(D) None of these

8. If $y = \sec^{-1} \left(\frac{x+1}{x-1} \right) + \sin^{-1} \left(\frac{x-1}{x+1} \right)$, then $\frac{dy}{dx} =$

(A) 0

(B) 1

(C) 2

(D) 3

9. If $y = \tan^{-1} \left[\frac{\sin x + \cos x}{\cos x - \sin x} \right], x \in \left(0, \frac{\pi}{4} \right)$ then $\frac{dy}{dx}$ is

(A) $\frac{1}{2}$

(B) $\frac{\pi}{4}$

(C) 0

(D) 1

10. If $y = \tan^{-1} (\cot x) + \cot^{-1} (\tan x)$, then $\frac{dy}{dx}$ is equal to -

(A) 1

(B) 0

(C) -1

(D) -2

11. The differential coefficient of $\sec^{-1} \left(\frac{1}{2x^2-1} \right)$ w.r.t. $\sqrt{1-x^2}$, where $x \in (0,1)$ is -

(A) $1/x^2$

(B) $2/x^3$

(C) $x/2$

(D) $2/x$

12. $\frac{d}{dx} \cot^{-1} \left(\frac{1+x}{1-x} \right)$ is equal to, if $x > -1$

- (A) $\frac{1}{1+x^2}$ (B) $\frac{1}{1-x^2}$ (C) $-\frac{1}{1+x^2}$ (D) $\frac{-1}{1-x^2}$

13. $\frac{d}{d\theta} \left(\tan^{-1} \left(\frac{1-\cos\theta}{\sin\theta} \right) \right)$ equals, if $-\pi < \theta < \pi$

- (A) $1/2$ (B) 1 (C) $\sec\theta$ (D) $\operatorname{cosec}\theta$

14. Prove that the function $y = \frac{\arcsin x}{\sqrt{1-x^2}}$ satisfies the relationship $(1-x^2)y' - xy = 1$ (where dash denotes derivative).

9. DERIVATIVE OF A FUNCTION AND ITS INVERSE FUNCTION :

If g is inverse of f , then

- (a) $g\{f(x)\} = x$ (b) $f\{g(x)\} = x$
 $g'\{f(x)\}f'(x) = 1$ $f'\{g(x)\}g'(x) = 1$

Illustration 16 : If g is inverse of f and $f'(x) = \frac{1}{1+x^n}$, then $g'(x)$ equals :-

- (A) $1+x^n$ (B) $1+[f(x)]^n$ (C) $1+[g(x)]^n$ (D) none of these

Solution : Since g is the inverse of f . Therefore

$$f(g(x)) = x \quad \text{for all } x$$

$$\Rightarrow \frac{d}{dx} f(g(x)) = 1 \quad \text{for all } x$$

$$\Rightarrow f'(g(x)) g'(x) = 1 \Rightarrow g'(x) = \frac{1}{f'(g(x))} = 1 + (g(x))^n \quad \text{Ans. (C)}$$

Do yourself -7 :

1. If g is inverse of f and $f(x) = 2x + \sin x$; then $g'(x)$ equals :

- (A) $-\frac{3}{x^2} + \frac{1}{\sqrt{1-x^2}}$ (B) $2 + \sin^{-1}x$ (C) $2 + \cos g(x)$ (D) $\frac{1}{2 + \cos(g(x))}$

2. Let $g(x) = f^{-1}(x)$ where $f(x) = x^3 + 3x$, then find $f^{-1}(4)$ or $g(4)$.

3. Let $g(x) = f^{-1}(x)$ where $f(x) = x + e^x$ then find $f^{-1}(1)$.

4. Let $f(x) = x \sin x \forall x \in \left(0, \frac{\pi}{2}\right)$ and g be the inverse of f in $\left(0, \frac{\pi}{2}\right)$, then $(\sqrt{3}\pi + 6)g'\left(\frac{\pi}{12}\right)$ is equal to

- (A) 6 (B) 12 (C) 4 (D) 3

5. If $f(x) = x + 3x^3 + 5x^5$ and $g = f^{-1}$, then find $g'(9)$ and $g''(9)$

6. Let $g(x)$ is the inverse function of a differentiable function $f(x)$ and $G(x) = -\frac{4}{g^2(x)}$. If

$$f(5) = 3 \text{ and } f'(5) = \frac{1}{125}, \text{ then } G'(3) \text{ is equal to}$$

10. HIGHER ORDER DERIVATIVES :

Let a function $y = f(x)$ be defined on an interval (a, b) . If $f(x)$ is differentiable function, then its derivative $f'(x)$ [or (dy/dx) or y'] is called the first derivative of y w.r.t. x . If $f'(x)$ is again differentiable function on (a, b) , then its derivative $f''(x)$ [or d^2y/dx^2 or y''] is called second derivative of y w.r.t. x .

Similarly, the 3rd order derivative of y w.r.to x , if it exists, is defined by $\frac{d^3y}{dx^3} = \frac{d}{dx} \left(\frac{d^2y}{dx^2} \right)$ and denoted by $f'''(x)$ or y''' and so on.

Note : If $x = f(\theta)$ and $y = g(\theta)$ where ' θ ' is a parameter then $\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta}$ & $\frac{d^2y}{dx^2} = \frac{d}{d\theta} \left(\frac{dy}{dx} \right) \bigg/ \frac{dx}{d\theta}$

In general $\frac{d^n y}{dx^n} = \frac{d}{d\theta} \left(\frac{d^{n-1} y}{dx^{n-1}} \right) \bigg/ \frac{dx}{d\theta}$.

11. ANALYSIS AND GRAPHS OF SOME INVERSE TRIGONOMETRIC FUNCTIONS :

$$(a) \quad y = f(x) = \sin^{-1} \left(\frac{2x}{1+x^2} \right) = \begin{cases} 2 \tan^{-1} x & |x| \leq 1 \\ \pi - 2 \tan^{-1} x & x > 1 \\ -(\pi + 2 \tan^{-1} x) & x < -1 \end{cases}$$

Important points :

- (i) Domain is $x \in \mathbb{R}$ & range is $\left[-\frac{\pi}{2}, \frac{\pi}{2} \right]$
- (ii) f is continuous for all x but not differentiable at $x = 1, -1$

$$(iii) \quad \frac{dy}{dx} = \begin{cases} \frac{2}{1+x^2} & \text{for } |x| < 1 \\ \text{non existent} & \text{for } |x| = 1 \\ \frac{-2}{1+x^2} & \text{for } |x| > 1 \end{cases}$$

- (iv) Increasing in $[-1, 1]$ & Decreasing in $(-\infty, -1], [1, \infty)$

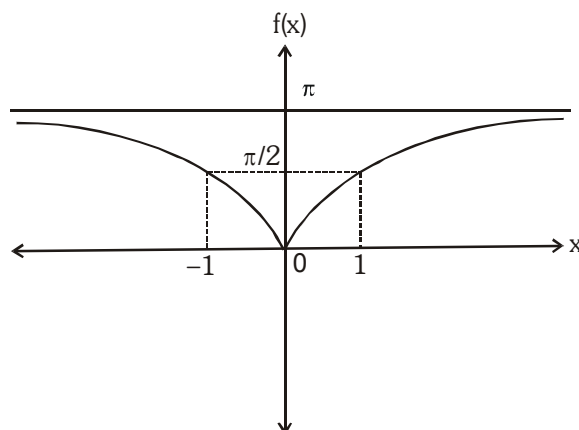
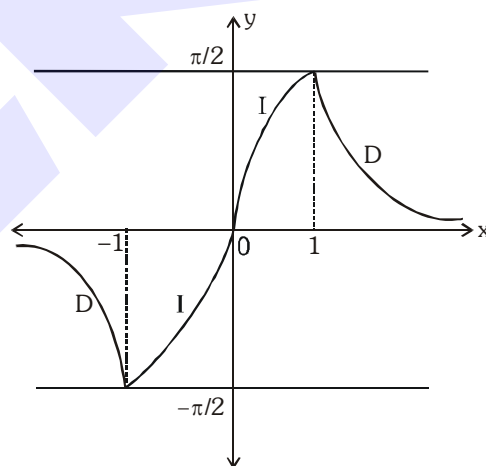
$$(b) \quad \text{Consider } y = f(x) = \cos^{-1} \left(\frac{1-x^2}{1+x^2} \right) = \begin{cases} 2 \tan^{-1} x & \text{if } x \geq 0 \\ -2 \tan^{-1} x & \text{if } x < 0 \end{cases}$$

Important points :

- (i) Domain is $x \in \mathbb{R}$ & range is $[0, \pi)$
- (ii) Continuous for all x but not differentiable at $x = 0$

$$(iii) \quad \frac{dy}{dx} = \begin{cases} \frac{2}{1+x^2} & \text{for } x > 0 \\ \text{non existent} & \text{for } x = 0 \\ -\frac{2}{1+x^2} & \text{for } x < 0 \end{cases}$$

- (iv) Increasing in $[0, \infty)$ & Decreasing in $(-\infty, 0]$



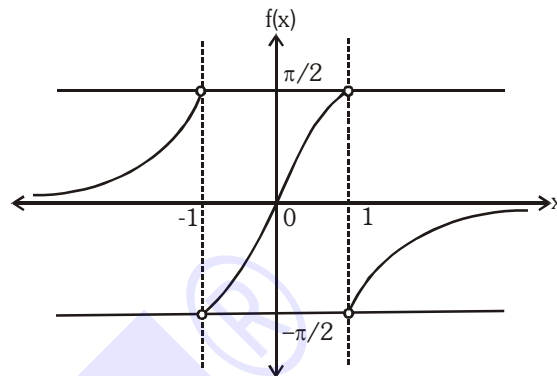
$$(c) \quad y = f(x) = \tan^{-1} \frac{2x}{1-x^2} = \begin{cases} 2 \tan^{-1} x & |x| < 1 \\ \pi + 2 \tan^{-1} x & x < -1 \\ -(\pi - 2 \tan^{-1} x) & x > 1 \end{cases}$$

Important points :

(i) Domain is $\mathbb{R} - \{1, -1\}$ & range is $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

$$(ii) \quad \frac{dy}{dx} = \begin{cases} \frac{2}{1+x^2} & |x| \neq 1 \\ \text{non existent} & |x| = 1 \end{cases}$$

(iii) It is bounded for all x



$$(d) \quad y = f(x) = \sin^{-1}(3x - 4x^3) = \begin{cases} -(\pi + 3 \sin^{-1} x) & \text{if } -1 \leq x < -\frac{1}{2} \\ 3 \sin^{-1} x & \text{if } -\frac{1}{2} \leq x \leq \frac{1}{2} \\ \pi - 3 \sin^{-1} x & \text{if } \frac{1}{2} < x \leq 1 \end{cases}$$

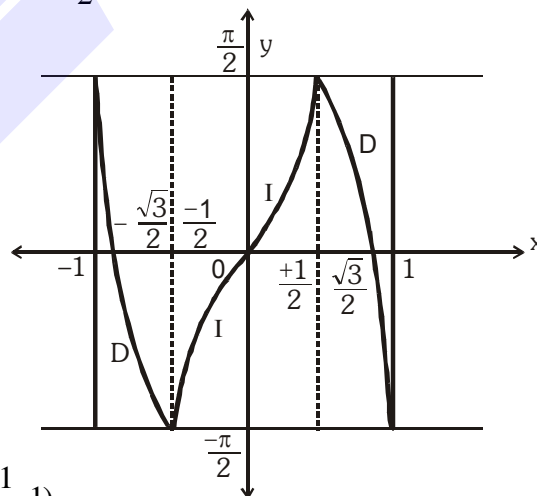
Important points :

(i) Domain is $x \in [-1, 1]$ & range is $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

(ii) Continuous everywhere in its domain

(iii) Not derivable at $x = -\frac{1}{2}, \frac{1}{2}, 1, -1$

$$(iv) \quad \frac{dy}{dx} = \begin{cases} \frac{3}{\sqrt{1-x^2}} & \text{if } x \in \left(-\frac{1}{2}, \frac{1}{2}\right) \\ -\frac{3}{\sqrt{1-x^2}} & \text{if } x \in \left(-1, -\frac{1}{2}\right) \cup \left(\frac{1}{2}, 1\right) \end{cases}$$



(v) Increasing in $\left[-\frac{1}{2}, \frac{1}{2}\right]$ and Decreasing in $\left[-1, -\frac{1}{2}\right], \left[\frac{1}{2}, 1\right]$

$$(e) \quad y = f(x) = \cos^{-1}(4x^3 - 3x) = \begin{cases} 3 \cos^{-1} x - 2\pi & \text{if } -1 \leq x < -\frac{1}{2} \\ 2\pi - 3 \cos^{-1} x & \text{if } -\frac{1}{2} \leq x \leq \frac{1}{2} \\ 3 \cos^{-1} x & \text{if } \frac{1}{2} < x \leq 1 \end{cases}$$

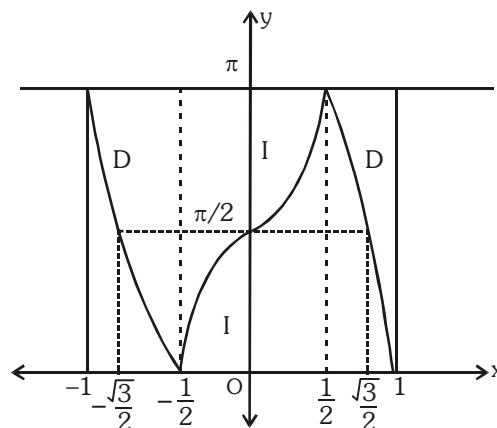
Important points :

- (i) Domain is $x \in [-1, 1]$ & range is $[0, \pi]$
- (ii) Continuous everywhere in its domain
- (iii) Not derivable at $x = -\frac{1}{2}, \frac{1}{2}, 1, -1$

$$(iv) \frac{dy}{dx} = \begin{cases} \frac{3}{\sqrt{1-x^2}} & \text{if } x \in \left(-\frac{1}{2}, \frac{1}{2}\right) \\ -\frac{3}{\sqrt{1-x^2}} & \text{if } x \in \left(-1, -\frac{1}{2}\right) \cup \left(\frac{1}{2}, 1\right) \end{cases}$$

(v) Increasing in $\left[-\frac{1}{2}, \frac{1}{2}\right]$ &

Decreasing in $\left[-1, -\frac{1}{2}\right], \left[\frac{1}{2}, 1\right]$



GENERAL NOTE :

Concavity is decided by the sign of 2nd derivative as :

$$\frac{d^2y}{dx^2} \geq 0 \Rightarrow \text{Concave upwards ; } \frac{d^2y}{dx^2} \leq 0 \Rightarrow \text{Concave downwards}$$

Illustration 17: Find the interval for which $f(x) = x^3 + x + 1$ is

- (i) concave upwards
- (ii) concave downwards.

Solution :

$$f(x) = x^3 + x + 1$$

$$f'(x) = 3x^2 + 1$$

$$f''(x) = 6x$$

$$(i) f''(x) = 6x \geq 0 \Rightarrow \text{Concave upwards}$$

$$\Rightarrow x \in [0, \infty)$$

$$(ii) f''(x) = 6x \leq 0 \Rightarrow \text{Concave downwards}$$

$$\Rightarrow x \in (-\infty, 0]$$

Ans.

Illustration 18: If $x = a(t + \sin t)$ and $y = a(1 - \cos t)$, find $\frac{d^2y}{dx^2}$.

Solution : Here $x = a(t + \sin t)$ and $y = a(1 - \cos t)$
Differentiating both sides w.r.t. t , we get :

$$\frac{dx}{dt} = a(1 + \cos t) \quad \text{and} \quad \frac{dy}{dt} = a(\sin t)$$

$$\therefore \frac{dy}{dx} = \frac{a \sin t}{a(1 + \cos t)} = \frac{2 \sin \frac{t}{2} \cdot \cos \frac{t}{2}}{2 \cos^2 \frac{t}{2}} = \tan\left(\frac{t}{2}\right)$$

Again differentiating both sides, we get,

$$\frac{d^2y}{dx^2} = \sec^2\left(\frac{t}{2}\right) \cdot \frac{1}{2} \cdot \frac{dt}{dx} = \frac{1}{2} \sec^2(t/2) \cdot \frac{1}{a(1 + \cos t)} = \frac{1}{2a} \cdot \frac{\sec^2\left(\frac{t}{2}\right)}{2 \cos^2 \frac{t}{2}}$$

$$\text{Hence,} \quad \frac{d^2y}{dx^2} = \frac{1}{4a} \sec^4\left(\frac{t}{2}\right)$$

Ans.

Illustration 19: $y = f(x)$ and $x = g(y)$ are inverse functions of each other then express $g'(y)$ and $g''(y)$ in terms of derivative of $f(x)$.

Solution : $\frac{dy}{dx} = f'(x)$ and $\frac{dx}{dy} = g'(y)$

$$\Rightarrow g'(y) = \frac{1}{f'(x)} \quad \dots\dots\dots(i)$$

Again differentiating w.r.t. to y

$$g''(y) = \frac{d}{dy} \left(\frac{1}{f'(x)} \right) = \frac{d}{dx} \left(\frac{1}{f'(x)} \right) \cdot \frac{dx}{dy} = -\frac{f''(x)}{(f'(x))^2} \cdot \left(\frac{1}{f'(x)} \right)$$

$$\Rightarrow g''(y) = -\frac{f''(x)}{(f'(x))^3} \quad \dots\dots\dots(ii)$$

$$\text{Which can also be remembered as} \quad \frac{d^2x}{dy^2} = -\frac{\frac{d^2y}{dx^2}}{\left(\frac{dy}{dx}\right)^3}$$

Ans.

Do yourself : 8

1. If $y = xe^{x^2}$ then find y'' .
2. Find y'' at $x = \pi/4$, if $y = x \tan x$.
3. Prove that the function $y = e^x \sin x$ satisfies the relationship $y'' - 2y' + 2y = 0$.
4. If $y = at^2 + 2bt + c$ and $t = ax^2 + 2bx + c$, then $\frac{d^3y}{dx^3}$ equals
(A) $24 a^2 (at + b)$ (B) $24 a (ax + b)^2$ (C) $24 a (at + b)^2$ (D) $24 a^2 (ax + b)$

12. DIFFERENTIATION OF DETERMINANTS :

If $F(x) = \begin{vmatrix} f(x) & g(x) & h(x) \\ l(x) & m(x) & n(x) \\ u(x) & v(x) & w(x) \end{vmatrix}$, where $f, g, h, l, m, n, u, v, w$ are differentiable functions of x then

$$F'(x) = \begin{vmatrix} f'(x) & g'(x) & h'(x) \\ l(x) & m(x) & n(x) \\ u(x) & v(x) & w(x) \end{vmatrix} + \begin{vmatrix} f(x) & g(x) & h(x) \\ l'(x) & m'(x) & n'(x) \\ u(x) & v(x) & w(x) \end{vmatrix} + \begin{vmatrix} f(x) & g(x) & h(x) \\ l(x) & m(x) & n(x) \\ u'(x) & v'(x) & w'(x) \end{vmatrix}$$

Note : Sometimes it is better to expand the determinant first & then differentiate.

Illustration 20: If $f(x) = \begin{vmatrix} x & x^2 & x^3 \\ 1 & 2x & 3x^2 \\ 0 & 2 & 6x \end{vmatrix}$, find $f'(x)$.

Solution : Here, $f(x) = \begin{vmatrix} x & x^2 & x^3 \\ 1 & 2x & 3x^2 \\ 0 & 2 & 6x \end{vmatrix}$

On differentiating, we get,

$$\Rightarrow f'(x) = \begin{vmatrix} \frac{d}{dx}(x) & \frac{d}{dx}(x^2) & \frac{d}{dx}(x^3) \\ 1 & 2x & 3x^2 \\ 0 & 2 & 6x \end{vmatrix} + \begin{vmatrix} x & x^2 & x^3 \\ \frac{d}{dx}(1) & \frac{d}{dx}(2x) & \frac{d}{dx}(3x^2) \\ 0 & 2 & 6x \end{vmatrix} + \begin{vmatrix} x & x^2 & x^3 \\ 1 & 2x & 3x^2 \\ \frac{d}{dx}(0) & \frac{d}{dx}(2) & \frac{d}{dx}(6x) \end{vmatrix}$$

$$\text{or } f'(x) = \begin{vmatrix} 1 & 2x & 3x^2 \\ 1 & 2x & 3x^2 \\ 0 & 2 & 6x \end{vmatrix} + \begin{vmatrix} x & x^2 & x^3 \\ 0 & 2 & 6x \\ 0 & 2 & 6x \end{vmatrix} + \begin{vmatrix} x & x^2 & x^3 \\ 1 & 2x & 3x^2 \\ 0 & 0 & 6 \end{vmatrix}$$

As we know if any two rows or columns are equal, then value of determinant is zero.

$$= 0 + 0 + \begin{vmatrix} x & x^2 & x^3 \\ 1 & 2x & 3x^2 \\ 0 & 0 & 6 \end{vmatrix} \quad \therefore f'(x) = 6(2x^2 - x^2)$$

Therefore, $f'(x) = 6x^2$

Ans.

Do yourself : 9

1. If $f(x) = \begin{vmatrix} e^x & x^2 \\ \ln x & \sin x \end{vmatrix}$, then find $f'(1)$.
2. If $f(x) = \begin{vmatrix} 2x & x^2 & x^3 \\ x^2 + 2x & 1 & 3x + 1 \\ 2x & 1 - 3x^2 & 5x \end{vmatrix}$, then find $f'(1)$.
3. If $f_n(x)$, $g_n(x)$, $h_n(x)$, $n = 1, 2, 3$ are polynomials in x such that $f_n(a) = g_n(a) = h_n(a)$, $n = 1, 2, 3$ and $F(x) = \begin{vmatrix} f_1(x) & f_2(x) & f_3(x) \\ g_1(x) & g_2(x) & g_3(x) \\ h_1(x) & h_2(x) & h_3(x) \end{vmatrix}$. Then $F'(a)$ is equal to
 (A) 0 (B) $f_1(a)g_2(a)h_3(a)$ (C) 1 (D) None of these
4. Let $f(x) = \begin{vmatrix} x^3 & \sin x & \cos x \\ 6 & -1 & 0 \\ p & p^2 & p^3 \end{vmatrix}$, where p is a constant. Then $\frac{d^3}{dx^3} \{f(x)\}$ at $x = 0$ is
 (A) p (B) $p + p^2$ (C) $p + p^3$ (D) Independent of p
5. $f(x) = \begin{vmatrix} x^3 & x^2 & 3x^2 \\ 1 & -6 & 4 \\ p & p^2 & p^3 \end{vmatrix}$, where p is a constant, then $\frac{d^3 f(x)}{dx^3}$ is
 (A) Proportional to x^2 (B) Proportional to x (C) Proportional to x^3 (D) A constant
6. If $y = \sin px$ and y_n is the n^{th} derivative of y , then $\begin{vmatrix} y & y_1 & y_2 \\ y_3 & y_4 & y_5 \\ y_6 & y_7 & y_8 \end{vmatrix}$ is equal to
 (A) 1 (B) 0 (C) -1 (D) None of these

13. L' HÔPITAL'S RULE :

- (a) This rule is applicable for the indeterminate forms of the type $\frac{0}{0}$, $\frac{\infty}{\infty}$. If the function $f(x)$ and $g(x)$ are differentiable in certain neighbourhood of the point 'a', except, may be, at the point 'a' itself and $g'(x) \neq 0$, and if

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x) = 0 \quad \text{or} \quad \lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x) = \infty,$$

$$\text{then} \quad \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

provided the limit $\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$ exists (L'Hôpital's rule). The point 'a' may be either finite or improper ($+\infty$ or $-\infty$).

- (b) Indeterminate forms of the type $0 \cdot \infty$ or $\infty - \infty$ are reduced to forms of the type $\frac{0}{0}$ or $\frac{\infty}{\infty}$ by algebraic transformations.
- (c) Indeterminate forms of the type 1^∞ , ∞^0 or 0^0 are reduced to forms of the type $0 \times \infty$ by taking logarithms or by the transformation $[f(x)]^{\phi(x)} = e^{\phi(x) \cdot \ln f(x)}$.

Illustration 21: Evaluate $\lim_{x \rightarrow 0} |x|^{\sin x}$

Solution :

$$\begin{aligned} \lim_{x \rightarrow 0} |x|^{\sin x} &= \lim_{x \rightarrow 0} e^{\sin x \log_e |x|} = e^{\lim_{x \rightarrow 0} \frac{\log_e |x|}{\csc x}} \\ &= e^{\lim_{x \rightarrow 0} \frac{1/x}{-\csc x \cot x}} \quad (\text{applying L'Hôpital's rule}) \\ &= e^{\lim_{x \rightarrow 0} \frac{\sin^2 x}{x \cos x}} = e^{\lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^2 \cdot \left(\frac{x}{\cos x} \right)} = e^{-(1)^2 \cdot (1)} = e^0 = 1 \end{aligned}$$

Ans.

Illustration 22: Solve $\lim_{x \rightarrow 0^+} \log_{\sin x} \sin 2x$.

Solution : Here $\lim_{x \rightarrow 0^+} \log_{\sin x} \sin 2x$

$$\begin{aligned} &= \lim_{x \rightarrow 0^+} \frac{\log \sin 2x}{\log \sin x} \quad \left(\frac{-\infty}{-\infty} \text{ form} \right) \\ &= \lim_{x \rightarrow 0^+} \frac{\frac{1}{\sin 2x} \cdot 2 \cos 2x}{\frac{1}{\sin x} \cdot \cos x} \quad \{ \text{applying L'Hôpital's rule} \} \\ &= \lim_{x \rightarrow 0^+} \frac{\left(\frac{2x}{\sin(2x)} \right) \cos 2x}{\left(\frac{x}{\sin x} \right) \cos x} = \lim_{x \rightarrow 0^+} \frac{\cos 2x}{\cos x} = 1 \end{aligned}$$

Ans.

Illustration 23: Evaluate $\lim_{n \rightarrow \infty} \left(\frac{e^n}{\pi} \right)^{1/n}$.

Solution : Here, $A = \lim_{n \rightarrow \infty} \left(\frac{e^n}{\pi} \right)^{1/n}$ (∞^0 form)

$$\begin{aligned} \therefore \log A &= \lim_{n \rightarrow \infty} \frac{1}{n} \log \left(\frac{e^n}{\pi} \right) = \lim_{n \rightarrow \infty} \frac{n \log e - \log \pi}{n} \left(\frac{\infty}{\infty} \text{ form} \right) \\ &= \lim_{n \rightarrow \infty} \frac{\log e - 0}{1} \quad \{ \text{applying L'Hôpital's rule} \} \\ \log A = 1 &\Rightarrow A = e^1 \text{ or } \lim_{n \rightarrow \infty} \left(\frac{e^n}{\pi} \right)^{1/n} = e \end{aligned}$$

Ans.

Do yourself : 10

- Using L'Hôpital's rule find (a) $\lim_{x \rightarrow 0} \frac{\tan x - x}{x^3}$ (b) $\lim_{x \rightarrow 0} \frac{e^x - x - 1}{x^2}$
- Using L'Hôpital's rule verify that : (a) $\lim_{x \rightarrow 0} \frac{\sin x - \tan x}{x^3} = -\frac{1}{2}$ (b) $\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} = 1$
- If f is differentiable in $(0, 6)$ & $f'(4) = 5$, then $\lim_{x \rightarrow 2} \frac{f(4) - f(x^2)}{2 - x} =$
(A) 5 (B) 5/4 (C) 10 (D) 20
- Let $f(x)$ be differentiable at $x = h$, then $\lim_{x \rightarrow h} \frac{(x+h)f(x) - 2hf(h)}{x-h}$ is equal to -
(A) $f(h) + 2hf'(h)$ (B) $2f(h) + hf'(h)$ (C) $hf(h) + 2f'(h)$ (D) $hf(h) - 2f'(h)$
- Evaluate the following limits using L'Hospital's Rule or otherwise :
 $\lim_{x \rightarrow 0} \frac{x^{6000} - (\sin x)^{6000}}{x^2 \cdot (\sin x)^{6000}}$
- If $\lim_{x \rightarrow 0} \frac{1 - \cos x \cdot \cos 2x \cdot \cos 3x \dots \cos nx}{x^2}$ has the value equal to 253, find the value of n (where $n \in \mathbb{N}$).
- Given a real valued function $f(x)$ as follows :
 $f(x) = \frac{x^2 + 2\cos x - 2}{x^4}$ for $x < 0$; $f(0) = \frac{1}{12}$ & $f(x) = \frac{\sin x - \ln(e^x \cos x)}{6x^2}$ for $x > 0$. Test the continuity and differentiability of $f(x)$ at $x = 0$.

Miscellaneous Illustrations :

Illustration 24: Find second order derivative of $y = \sin x$ with respect to $z = e^x$.

Solution : $\frac{dy}{dz} = \frac{dy/dx}{dz/dx} = \frac{\cos x}{e^x}$

$$\Rightarrow \frac{d^2y}{dz^2} = \frac{d}{dx} \left(\frac{\cos x}{e^x} \right) \cdot \frac{dx}{dz} = \frac{-e^x \sin x - \cos x e^x}{(e^x)^2} \cdot \frac{1}{e^x}$$

$$\Rightarrow \frac{d^2y}{dz^2} = -\frac{(\sin x + \cos x)}{e^{2x}}$$

Ans.

Illustration 25: Let a function f satisfies $f\left(\frac{x+y}{2}\right) = \frac{f(x)+f(y)}{2} \forall x, y \in \mathbb{R}$ and $f'(0) = a, f(0) = b$, then find $f(x)$ hence find $f''(x)$.

Solution :
$$f\left(\frac{x+y}{2}\right) = \frac{f(x)+f(y)}{2}$$

Diff. w.r.t. 'x'

$$f'\left(\frac{x+y}{2}\right) \cdot \frac{1}{2} = \frac{1}{2} f'(x) \quad \left(\because x \text{ \& } y \text{ are independent to each other, } \therefore \frac{dy}{dx} = 0 \right)$$

$$f'\left(\frac{x+y}{2}\right) = f'(x)$$

Let $x = 0$ & $y = x$ $f'\left(\frac{x}{2}\right) = f'(0) = a$

$$\Rightarrow f'(x) = a$$

On integrating, we get $f(x) = ax + b$ ($\because f(0) = b$)

$$\Rightarrow f''(x) = 0$$

Illustration 26: Prove that $\frac{1}{2^2} \sec^2 \frac{x}{2} + \frac{1}{2^4} \sec^2 \frac{x}{2^2} + \frac{1}{2^6} \sec^2 \frac{x}{2^3} + \dots \infty = \operatorname{cosec}^2 x - \frac{1}{x^2}$

Solution : Let $\cos \frac{x}{2} \cdot \cos \frac{x}{2^2} \cdot \cos \frac{x}{2^3} \dots \infty$

$$= \lim_{n \rightarrow \infty} \cos \frac{x}{2} \cdot \cos \frac{x}{2^2} \cdot \cos \frac{x}{2^3} \dots \cos \frac{x}{2^n}$$

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{\sin x}{2^n \sin \frac{x}{2^n}} = \frac{\sin x}{x} \Rightarrow \cos \frac{x}{2} \cdot \cos \frac{x}{2^2} \cos \frac{x}{2^3} \dots \infty = \frac{\sin x}{x}$$

$$\Rightarrow \ln\left(\cos \frac{x}{2}\right) + \ln\left(\cos \frac{x}{2^2}\right) + \ln\left(\cos \frac{x}{2^3}\right) + \dots \infty = \ln \sin x - \ln x$$

Diff. w.r.t. x

$$-\left(\frac{1}{2} \tan \frac{x}{2} + \frac{1}{2^2} \tan \frac{x}{2^2} + \dots \infty\right) = \cot x - \frac{1}{x}$$

Diff. w.r.t. x again

$$\frac{1}{2^2} \sec^2 \frac{x}{2} + \frac{1}{2^4} \sec^2 \frac{x}{2^2} + \frac{1}{2^6} \sec^2 \frac{x}{2^3} + \dots \infty = \operatorname{cosec}^2 x - \frac{1}{x^2}$$

Hence proved